

Rishabh Tiwari

Senior Software Engineer – Python | AWS Cloud | AI/ML

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PROFESSIONAL SUMMARY

- **Senior Software Engineer with 8+ years of experience** designing and building **cloud-native systems, real-time automation platforms, distributed data pipelines, and AI-powered applications** using **Python, AWS, and modern DevOps** practices. Proven expertise in **serverless architectures, microservices, ETL/ELT pipelines, MLOps workflows, and LLM-based systems**. Skilled in architecting **scalable, secure, and high-performance solutions** that drive automation, reliability, and cost efficiency across large enterprise environments.
- Hands-on experience across **AWS (Lambda, Step Functions, DynamoDB, Glue, S3, EKS, EC2, EventBridge, API Gateway, Kinesis)** and limited **GCP** exposure. Proficient in **backend engineering with Python (FastAPI, Flask, Django), containerization (Docker, Kubernetes), and IaC (Terraform)** with **CI/CD automation and observability stacks**. Experienced in deploying and operationalizing **Large Language Models (GPT-4, Claude, Llama)** through **RAG architectures, vector search, embeddings, and secure multimodal AI workflows**, with a strong focus on **scalability, performance optimization, security, governance, and cloud cost optimization**.

CORE TECHNICAL SKILLS

- ➔ **Programming:** Python (expert), SQL, Bash, C++, Go, REST APIs, Async Python, Multithreading
- ➔ **AWS Cloud (Primary):** Lambda, API Gateway, DynamoDB, S3, Step Functions, Glue, EC2, EKS, ECS, CloudWatch, CloudTrail, IAM, KMS, RDS, SNS, SQS, EventBridge, CDK, CloudFormation
- ➔ **AI/ML + LLM:** Scikit-learn, PyTorch, TensorFlow, SageMaker, MLflow, Feature Engineering, RAG, LangChain, LlamaIndex, Pinecone, FAISS, ChromaDB, Embeddings, Prompt Engineering
- ➔ **Data Engineering & ETL:** AWS Glue, PySpark, Airflow, Apache Beam, Kinesis, Kafka, Snowflake, dbt, Data Lakehouse Design
- ➔ **DevOps & Infrastructure:** Docker, Kubernetes (EKS/GKE), Terraform, Jenkins, GitHub Actions, GitLab CI, Helm, ArgoCD, Logging/Monitoring (Grafana, Prometheus, ELK)
- ➔ **Databases:** PostgreSQL, MySQL, DynamoDB, MongoDB, Redis, Redshift, Snowflake, Elasticsearch

CERTIFICATIONS

- AWS Solutions Architect – Professional (2025)
- AWS Developer Associate (2025)
- AWS AI Practitioner (2025)

PROFESSIONAL EXPERIENCE

WCG – Senior Software Engineer | May 2025 – Present

Domain: Cloud Data Engineering, Backend Engineering, Automation

Tech: Python, AWS, Azure, FastAPI, REST, Postgres, Jenkins, Dataflow

Key Contributions & Responsibilities:

- **Engineered large-scale Python microservices** to replace legacy batch systems, reducing execution time and operational overhead across workflows.
- **Designed and deployed serverless architectures** on **AWS Lambda, Step Functions, EventBridge, and API Gateway** to automate complex business processes.
- **Developed FastAPI-based backend APIs** for ingestion, validation, transformation, and orchestration across distributed systems.
- **Built internal Python automation frameworks** for resource provisioning, configuration drift detection, cost tracking, and audit logging.
- **Created Python SDKs and CLI tools** enabling developers to safely interact with internal systems, automate recurring tasks, and streamline deployments.
- **Implemented LLM-powered solutions** for document summarization, intelligent search, and query answering using **RAG pipelines, embeddings, and vector databases**.
- **Built a secure multi-tenant RAG microservice** on **AWS EKS**, integrating **Pinecone/FAISS** with **Secrets Manager** for isolated credential management.
- **Designed automated compliance pipelines** to monitor **PII access**, enforce **IAM least-privilege**, and validate configuration integrity.
- **Defined AWS resources using Terraform IaC** (IAM, networking, Lambda, API Gateway, S3), enabling reproducible and version-controlled deployments.
- **Implemented scalable CI/CD pipelines in GitHub Actions** to automate testing, packaging, container builds, and infrastructure deployments.
- **Developed telemetry and monitoring agents** to stream logs and metrics to **CloudWatch, Elasticsearch, and Grafana** for real-time observability.
- **Led cloud modernization and AI-driven initiatives**, migrating monoliths to **EKS microservices**, integrating anomaly detection, compliance reporting, and secure API management for clinical workflows.

Brudite – Senior Software Engineer | Jan 2025 – May 2025

Domain: AI/ML Engineering, AWS, Cloud Observability, Backend Systems

Tech: Python, FastAPI, AWS Lambda, Step Functions, DynamoDB, S3, Sagemaker, MLflow, Grafana, Prometheus

Key Contributions & Responsibilities:

- **Built ML-driven anomaly detection and auto-remediation workflows** using SageMaker, Lambda, and Step Functions, enabling intelligent event orchestration and automation.
- **Designed and deployed FastAPI microservices** for model prediction serving, inference pipelines, and automation triggers with integrated health checks and observability.
- **Implemented RAG-based architectures** for log summarization and operational analytics using embeddings, vector databases, and LLM agents.
- **Automated model drift detection, retraining, and deployment** using EventBridge, GitHub Actions, and Terraform for consistent, versioned ML delivery.
- **Orchestrated scalable containerized ML workloads** on ECR and ECS, ensuring reproducible inference pipelines and resilient deployments.
- **Integrated MLflow** for experiment tracking, versioning, and explainability to maintain robust ML governance and auditability.

- **Developed Python monitoring agents and self-healing mechanisms**, leveraging Prometheus/Grafana dashboards to collect metrics, alert, and automatically respond to model degradation or inference anomalies.
- **Collaborated with DevOps/security teams** to enforce IAM policies, encryption, secret rotation, and implemented observability dashboards, audit frameworks, and cost-optimization audits across ML pipelines.

IRIS – Software Engineer | May 2023 – Dec 2024

Domain: AWS Automation, Backend Engineering, DevOps

Tech: Python, Bash, AWS (EC2, RDS, ECS/EKS, Lambda), Jenkins, Docker, Kubernetes

Responsibilities:

- **Built Python and Bash automation scripts** for AWS RDS backups, restores, scaling, and health checks, improving reliability and reducing manual intervention.
- **Developed event-driven automation** using AWS Lambda and EventBridge for self-healing infrastructure during spikes and failures.
- **Designed REST APIs in Python** to automate provisioning, enforce policies, and standardize multi-account infrastructure operations.
- **Containerized workloads** with Docker, ECS, and Kubernetes, ensuring high availability, scalability, and fault tolerance.
- **Engineered Terraform-based IaC frameworks** and CI/CD pipelines (Jenkins, GitHub Actions) for consistent provisioning, builds, testing, packaging, and multi-environment deployments.
- **Optimized security, governance, cost, and data pipelines** by refining IAM roles, automating secrets rotation (KMS/Secrets Manager), enforcing least-privilege policies, implementing cleanup/right-sizing scripts, and modernizing ETL workflows to dbt + Snowflake with automated CI/CD.

ConsultAdd Services – Software Engineer Level 3 | July 2018 – May 2023

Domain: Cloud Engineering, Backend Systems, Security Automation, EdTech Platforms

Tech: Python, Django, Flask, Docker, Kubernetes, GCP (minimal), Terraform, SQL

Responsibilities:

Cloud & Backend Engineering:

- Designed and developed **Python-based backend systems** using **Django, Flask, and DRF**, powering high-traffic applications across retail, logistics, and security domains.
- Built **modular REST APIs** for authentication, workflow automation, and device integration, enhanced with **async workers (Celery + Pub/Sub/SQS)** for parallel task execution.
- Containerized and deployed **microservices on GKE/EKS** using **Docker**, implementing Kubernetes best practices such as probes, HPA, and rolling updates for scalability.
- Developed **file-processing and automation services** for log parsing, metadata extraction, and multi-service orchestration using Python.
- Implemented **secure backend architectures** with **JWT-based auth, RBAC**, and caching strategies, improving system reliability, performance, and data protection.

Security & DDoS Protection

- Engineered a **cloud-native DDoS detection and mitigation platform** using **Python, Kubernetes, Cloud Armor, Elasticsearch, and Kibana**, enhancing system resiliency and response time during attacks.
- Developed advanced **packet inspection and analysis utilities** with **tcpdump** and **Scapy** to identify malformed packets, anomalies, and emerging threat patterns in real time.

- Automated incident detection and mitigation workflows through **event-driven pipelines (Pub/Sub, SQS)** that dynamically triggered firewall updates, alerts, and protection policies.
- Built secure **monitoring and defense frameworks** integrating **Grafana, Kibana, encryption, anomaly scoring**, and retention controls, collaborating with security teams to enforce firewall and VPC isolation.

Data Pipelines & Integrations

- Developed **Python-based integration services** for inventory, logistics, and device data flows using **REST APIs** and **cloud event systems**, ensuring seamless interoperability across distributed platforms.
- Automated warehouse and inventory operations with **Django Admin** and backend logic to enable real-time tracking of collections, shipping, returns, and quality control processes.
- Built scalable ingestion and automation pipelines for **CSV/JSON/XML data** using **Python** and **Cloud Functions**, implementing transformations, validations, retries, DLQs, and integrations with **Bitbucket, Jira, and Slack** for monitoring and reporting.

Cloud Infrastructure & DevOps

- Provisioned and managed multi-cloud infrastructure across **AWS and GCP** using **Terraform**, automating setup of VPCs, IAM, Kubernetes clusters, databases, and load balancers for scalable deployments.
- Automated **CI/CD pipelines** with Cloud Build, GitHub Actions, and GitLab CI, and containerized Python applications, ensuring secure, versioned, and compliant deployments with minimal manual intervention.
- Implemented enterprise-grade **security and monitoring** via network isolation, least-privilege IAM roles, KMS/Secrets Manager encryption, and Grafana and ELK dashboards for real-time visibility, cost control, and compliance automation.

EdTech Platform Development

- Developed **Python-based integration and automation services** for inventory, logistics, and device data flows using **REST APIs, Django Admin, and cloud event systems**, enabling seamless real-time operations and visibility.
- Built **scalable ingestion and automation pipelines** for **CSV/JSON/XML data** using **Python** and **Cloud Functions**, automating transformations, validations, and device lifecycle workflows with **checkpointing, retries, DLQs**, and integrations with **Bitbucket, Jira, Slack, and Google APIs** for observability and reporting.

Localization Engineering for Apple

- Developed and automated **Python-based localization workflows** for **Apple iOS/macOS apps**, validating **multilingual content (EN/DE/JA/AR)** and **UI strings** through **CI/CD-integrated pipelines (Pub/Sub, Cloud Run, device orchestration)**, **containerized services on GKE**, and **internal dashboards/APIs** for coverage, build quality, and region-specific defects, collaborating with **global QA teams** to accelerate releases and ensure high-quality international builds.

EDUCATION

Bachelor's in Computer Science Engineering

Rajasthan Technical University (2014 – 2018)